

HAT-P-36b

R-0789

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_250503 · created 2026-07-28T07:23:12+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460798.9092 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1571 +/- 0.0081
Transit depth (Rp/Rs)^2	2.47 +/- 0.25 [%]
Orbital Inclination (inc)	84.72 +/- 2.31
Airmass coefficient 1 (a1)	1.027 +/- 0.012
Airmass coefficient 2 (a2)	-0.0108 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	1.22 %
Best Comparison Star	#3 - [147, 453]
Optimal Method	PSF photometry
Transit Duration (day)	0.0929 +/- 0.0061

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1571 ± 0.0081 vs 0.126 (deviation 3.84σ)
Duration fit vs published	0.0929 d vs 0.0925 d (deviation 0.07σ)
Mid-time O-C	-1.8 min
Depth SNR	19.4
Residual scatter	1.22 %

Light curve

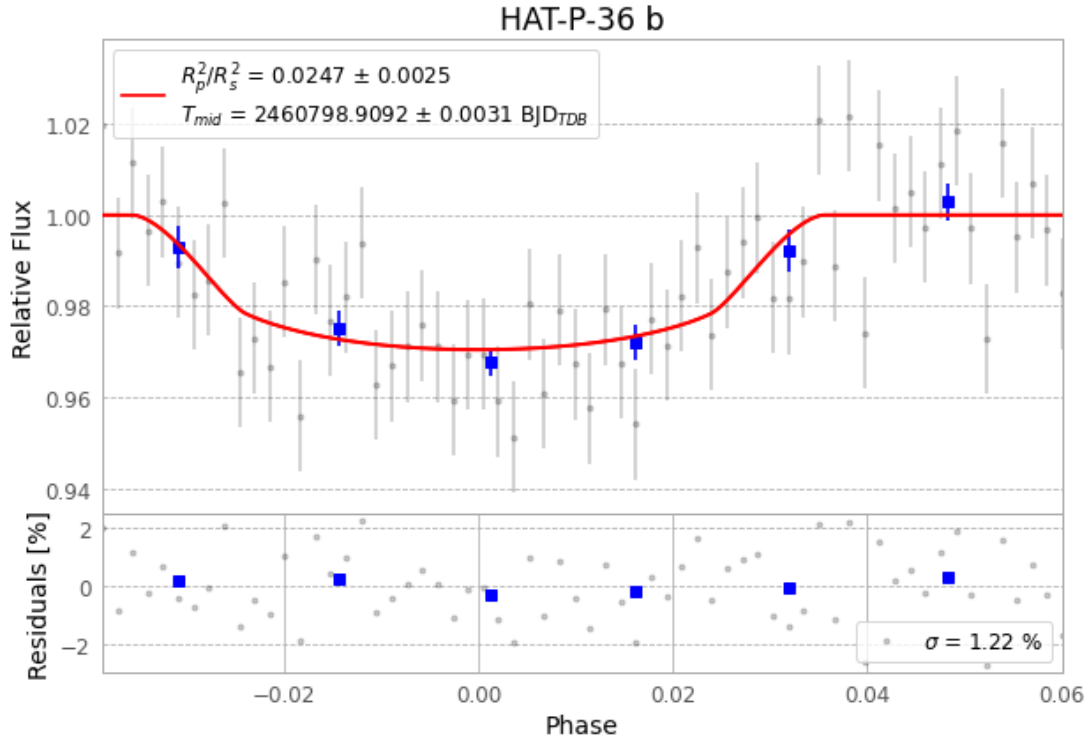


Figure 1 — FinalLightCurve_HAT-P-36 b_2025-05-03.png (SHA-256 4bc1abdacf4b9baa...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 306], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2a46519c56c0207a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

64 FITS frames (42 MB), HATP-36250503083015.FITS → HATP-36250503113916.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-250503.log (deaadede262f...), FinalLightCurve_HAT-P-36 b_2025-05-03.png (4bc1abdacf4b...), FinalLightCurve_HAT-P-36 b_2025-05-03.csv (344e8fd57b37...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 07:23Z.